FBISE Math Class 9 Test

Chapter 11 Basic Statistics

Sec A: MCQs

$\frac{15+20}{2} = 17.5$

13. Class mark of interval (15 – 20) is:	15	17.5	20	35
14. The probability of an event CANNOT be:	0	0.5	1	2
15. A boy scored 3 goals in 9 matches. His relative frequency of scoring goals is:	9	3	$\frac{1}{2}$	$\frac{1}{3}$

13. The median of 17, 27, 37, 47, 57, 67, 77 is:	47	17	77	0
14. If a fair die is rolled once, what is the probability of getting 2?	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{5}$
15. Which is a possible **Valid** relative frequency value:	−0.5	1.2	0.75	2

xiii.	The mean of 11 numbers is 7. One of the numbers 13 is deleted. What is the mean of the remaining 10 numbers?	7.7	6.4	6.0	5.8
xiv.	What is the probability of picking a king from well-shuffled 52 playing cards?	$\frac{1}{52}$	$\frac{1}{13}$	$\frac{4}{13}$	$\frac{1}{26}$
xv.	A fair coin is tossed twice, then the frequency of appearing head twice is:	$\frac{1}{4}$	$\frac{1}{3}$	$\frac{1}{2}$	$\frac{3}{4}$

13.	The median of the given data is: *77, 79, 82, 86, 90, 92, 93* دیے گئے مواد کا وسطانیہ کیا ہوگا؟	79	82	86	88
14.	What is the probability of getting Tail, if a fair coin is tossed once? اگر ایک متناسب سکہ اچھالا جائے تو 'ٹیل' آنے کا کتنا امکان ہے؟	0	$\frac{1}{2}$	1	2
15.	A die was rolled 50 times, and the number 4 appeared 15 times. What is the relative frequency of rolling a 4? لڈو کا ایک دانہ 50 مرتبہ پھینکنے پر 15 مرتبہ نمبر 4 آیا۔ نمبر 4 کی ریلیٹو فریکونسی کیا ہوگی؟	0.15	0.30	0.20	0.25

13.	A company surveyed the number of products purchased in a month: {2, 3, 3, 2, 4, 5, 3, 6}. What is the mode of the data? ایک کمپنی نے پراڈکٹس کی خرید کا سروے کیا۔ جس میں درج شمار سامنے آئے۔ {2, 3, 3, 2, 4, 5, 3, 6}اس مواد کا عادہ معلوم کریں۔	3	4	5	6
14.	A bag contains 4 blue balls and 6 red balls. If one ball is drawn at random, what is the probability that it is **NOT** blue? ایک بیگ میں 4 نیلی اور 6 سرخ گیندیں ہیں۔ اگر ایک گیند نکالی جائے تو اس بات کا کتنا امکان ہے کہ نکالی جانے والی گیند نیلی **نہیں** ہے؟	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{6}{5}$
15.	In 50 trials, an event occurs 10 times. What is the relative frequency of the event? 50 کوششوں میں ایک واقعہ 10 مرتبہ وقوع پذیر ہوتا ہے۔ اس واقعہ کی ریلیٹو فریکونسی کتنی ہے؟	$\frac{1}{5}$	$\frac{1}{10}$	$\frac{1}{2}$	$\frac{1}{50}$

13.	Median of 5, 7, 8, 10, 12 is: 5, 7, 8, 10, 12 کا وسطانیہ (Median) کیا ہے؟	8	9	10	7
14.	Probability of drawing a heart from a deck of 52 cards is: 52 کارڈز کے سیٹ سے ہارٹ نکالنے کا امکان کیا ہے؟	$\frac{1}{4}$	$\frac{1}{13}$	$\frac{1}{8}$	$\frac{1}{2}$
15.	Relative frequency of 8 successes out of 40 trials is: 40 کوششوں میں 8 کامیابیوں کی ریلیٹو فریکونسی کیا ہے؟	0.1	0.2	0.4	0.8

$\frac{10+15}{2} = \frac{25}{2} = 12.5$

1. Encircle the correct option in the following.

i. Which of the following is a class mark of the interval (10 – 15)?
(a) 10 (b) 12.5 (c) 15 (d) 16

4, 5, 6, 7

ii. What is size of class interval (4 – 7)? ⇒ 3.5 — 7.5
(a) 4 (b) 5 (c) 6 (d) 7

iii. Which of the following is chart of adjacent rectangles?
(a) bar graph (b) Frequency polygon (c) histogram (d) ogive

Bar

iv. Which of the following is measure of central tendency?
(a) variance (b) standard deviation (c) range (d) arithmetic mean

v. Which of the following is a formula of arithmetic mean for grouped data?
(a) $\frac{\Sigma x}{n}$ (b) $l + \frac{h}{f}\left(\frac{n}{2} - c\right)$ (c) $\frac{\Sigma fx}{\Sigma f}$ (d) $l + \frac{(f_m - f_1) \times h}{2f_m - f_1 - f_2}$

histo

vi. If arithmetic mean of 25 values is 10, then what is sum of values?
(a) 250 (b) 125 (c) 25 (d) 2.5

vii. What is median of the data 4, 3, 0, 2, 1? ⇒ 0, 1, 2, 3, 4
(a) 0 (b) 2 (c) 3 (d) 4

Mean = Sum / Nos.
10 (25)

viii. Which measure of central tendency can have more than one value?
(a) weighted mean (b) median (c) mode (d) arithmetic mean

ix. The probability of getting M in 'MUHAMMAD' is
(a) $\frac{1}{8}$ (b) $\frac{3}{8}$ (c) $\frac{3}{5}$ (d) none of these

x. Probability of picking an ace from a well shuffled pack of 52 playing cards is $\frac{4}{52}$
(a) $\frac{1}{52}$ (b) $\frac{1}{13}$ (c) $\frac{4}{13}$ (d) none of these

xi. A normal fair die is rolled 6000 times. The expected number of 5 is
(a) 100 (b) 5000 (c) 1000 (d) none of these

xii. A fair coin is tossed 500 times. Expected number of tails is
(a) 100 (b) 250 (c) 1000 (d) none of these

xiii. in a group of 5 people, 4 like peach juice. The expected no. of people in a population of 1200 who like peach juice is
(a) 240 (b) 600 (c) 960 (d) none of these

xiv. How many times Haani should toss a fair coin if he expects to get 100 tails

(a) 100 (b) 200 (c) 1000 (d) none of these

xv. An event which can never happen is called

(a) sure event (b) certain event (c) possible event (d) impossible event

xvi. Two such events whose probabilities are ½ each are called

(a) likely (b) equally likely (c) unlikely (d) none of these

xvii. The probability of a certain event is

(a) 0 (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 1

xviii. The probability of an event can take the value

(a) 0 (b) 1 (c) $\frac{1}{2}$ (d) all of these

xix. The probability of an event cannot take the value

(a) 0 (b) 1 (c) 2 (d) all of these

xx. The probability of an impossible event is

(a) 0 (b) 1 (c) 1/2 (d) all of these

Sec B: Short Questions

In a bag with 10 balls, there are 7 black and 3 white balls. If one ball is selected at random from the bag, calculate: a. The probability of selecting a white ball. b. The probability of selecting a black ball c. The sum of all the probability	04

The monthly incomes (in thousand rupees) of five workers are:

20, 22, 25, 90, 23.

Calculate the mean and median. Which measure better represents the income level? Give reason.

Question	Marks
In a coin toss experiment, Head is obtained 15 times in 50 trials. Calculate the relative frequencies of Head and Tails.	2+2
The given data shows the distribution of weights (in kg) of 30 bags of rice. **Calculate mean weight of** the rice bags.	04

$C-I$	05-09	10-14	15-19	20-24	25-29
f	5	7	10	4	4

Question	Marks
A number is chosen at random from a set of whole numbers from 1 to 50. Calculate the probability: **a.** Chosen number is not a perfect square. **b.** Chosen number is perfect square.	2+2
The populations (in thousands) of nine towns are: 24, 18, 30, 22, 26, 26, 26. 32, 36. Find Mode Median Mean	1+1 +2
A decagonal die labeled 4,4,4,4,5,5,6,7,8,8 is rolled once. Find the probability of an odd number, an even number, and a factor of 12.	4
A fair die is rolled 75 times and 5 appears up 20 times, what is the relative frequency of appearing any number up except 5.	4

In a 50-over cricket match, average runs scored by Pakistani team for different sessions of the innings is given below: **4**

The score in

01 to 10 overs: 12 runs per over,

11 to 35 overs: 06 runs per over,

36 to 50 overs: 13 runs per over.

Find average runs scored by the team in an innings.

A bag contains 5 red, 3 blue and 2 green balls. One ball is drawn at random from the bag. Calculate the probability that:

ایک بیگ میں 5 سرخ، 3 نیلی اور 2 سبز گیندیں ہیں۔ امکانات معلوم کریں کہ:

a The drawn ball is red. نکالی جانے والی گیند سرخ ہو **04**

b The drawn ball is **NOT** blue. نکالی جانے والی گیند نیلی نہ ہو

The given data shows the distribution of weights (in kg) of 30 bags of rice. **Calculate mean weight** of the rice bags

C-I	*05 – 09*	*10 – 14*	*15 – 19*	*20 – 24*	*25 – 29*
f	5	7	10	4	4

درج جدول میں چاول کے تھیلوں کے اوزان کی تقسیم دی گئی ہے۔ ان کا اوسط وزن معلوم کریں۔ **04**

A fair six-sided die is rolled 60 times. Calculate the expected frequency of

لڈو کا ایک متناسب دانہ 60 مرتبہ پھینکا گیا۔ درج شدہ ریلیٹو فریکونسی معلوم کریں:

a Rolling an even number side. ایک جفت عدد آنا

b Rolling a prime number side. ایک مفرد عدد آنا

04

A fair die is thrown once. Find the probability that face on the die is:

لڈو کا ایک متناسب دانہ ایک مرتبہ پھینکا گیا۔ درج شدہ کے امکانات معلوم کریں:

a A Prime number. ایک مفرد عدد آنا

b A Multiple of 3. عدد 3 کے اضعاف

04

The given table shows distribution of daily wages in US dollars of 50 overseas employees. **Calculate average wage** of the employees.

C-I	*10–14*	*15–19*	*20–24*	*25–29*	*30–34*
f	7	20	16	3	4

درج جدول میں امریکی ڈالروں میں 50 ملازمین کی تنخواہیں دی گئی ہیں۔ ان کی اوسط تنخواہ معلوم کریں۔

04

A fair coin is tossed 100 times. Calculate:

ایک متناسب سکہ 100 مرتبہ اچھالا گیا۔ معلوم کریں:

a Expected frequency of obtaining number of Heads. ہیڈ آنے کی متوقع فریکونسی

b Expected frequency of obtaining number of Tails. ٹیل آنے کی متوقع فریکونسی

04

Find the probability of getting a prime number when a fair die is rolled. ایک متناسب لڈو کا دانہ (fair die) پھینکنے پر مفرد عدد (prime number) حاصل ہونے کا امکان معلوم کریں۔	04
In 60 trials, event A occurred 12 times. Estimate probability and express as a fraction and decimal. 60 تجربات میں واقعہ A بارہ مرتبہ پیش آیا۔ اس کا امکان معلوم کریں اور اسے کسر (fraction) اور اعشاریہ (decimal) میں ظاہر کریں۔	04
Find the mean, mode and median of the data: 3, 4, 6, 6, 6, 7, 7, 8, 9. درج اعداد کا عادہ، اوسط اور وسطانیہ معلوم کریں: 3, 4, 6, 6, 6, 7, 7, 8, 9	04

Sec C: Long Questions

Consider the data and find the mean. **4+4**

CI	140-150	150-160	160-170	170-180	180-190
f	6	9	13	7	5

The daily consumption of electricity units of 35 factory machines is recorded as in given table. **08**

Calculate mode of the data.

C – I	20-24	25-29	30-34	35-39	40-44
f	8	12	2	10	3

The given table shows distribution of marks obtained by 50 students in a mathematics test: **08**

Calculate the median marks.

C-I	01 – 10	11 – 20	21 – 30	31 – 40	41 – 50
f	5	10	15	12	8

درج جدول میں ریاضی کے ٹیسٹ میں 50 طلبہ کے حاصل کردہ نمبروں کی تفصیل ہے۔ اس مواد کا وسطانیہ معلوم کریں۔

The grouped data for a company's monthly expense (in million rupees) is given as:	8

C-I	140 – 149	150 – 159	160 – 169	170 – 179
f	3	7	5	9

Calculate the median and mode expense for 24 months.

The given table shows distribution of marks obtained by 50 students in a mathematics test:

Calculate the median marks.

C-I	*01 – 10*	*11 – 20*	*21 – 30*	*31 – 40*	*41 – 50*
f	5	10	15	12	8

درج جدول میں ریاضی کے ٹیسٹ میں 50 طلبہ کے حاصل کردہ نمبروں کی تفصیل ہے۔ اس مواد کا وسطانیہ معلوم کریں۔

08

The daily consumption of electricity units of 35 factory machines is recorded as in given table.

Calculate mode of the data.

C-I	*20–24*	*25–29*	*30–34*	*35–39*	*40–44*
f	8	12	2	10	3

جدول میں ایک فیکٹری کی 35 مشینوں پر بجلی کے یونٹس کی کھپت دی گئی ہے۔ مواد کا عادہ معلوم کریں۔

08

A factory produces 200 bulbs daily. Their lifetimes are grouped in a frequency table. Construct a histogram. Find modal class and modal frequency.

Lifetime (in hours)	Frequency
300 – 400	10
400 – 500	30
500 – 600	60
600 – 700	70
700 – 800	30

08

ایک فیکٹری روزانہ 200 بلب بناتی ہے۔ ان کی لائف ٹائم (lifetime) کو ایک frequency table میں دیا گیا ہے۔

(i) ایک Histogram بنائیں۔

(ii) موڈل کلاس اور موڈل فریکونسی معلوم کریں۔

A box contains 2 red, 4 blue, 6 green and 8 yellow balls. A ball is drawn at random. Find the probability of each color.

ایک ڈبے میں 2 سرخ، 4 نیلے، 6 سبز اور 8 پیلے گیند ہیں۔ ہر رنگ کی گیند کے نکلنے کا امکان معلوم کریں۔

08